INYOUNG · CLASS 5

Teach Brian the Basics — and Grad School, Off the Record

First hour: things you already know cold, explained to someone who doesn't. Second hour: your actual life as a grad student.

PART 1 · 1 of 3

Pick Three

석사 1 년차 수업에서 다 배운 것들이에요 . Brian 은 하나도 몰라요 — 그게 오늘의 재료예요 .

1 **How a neuron fires**
Resting potential, threshold, the spike — and why it's all-or-nothing.
Brian asks: Is it like a light switch? / What's actually moving?

2 **The synapse**
Where one neuron talks to the next, and why the gap matters.
Brian asks: Why not just touch? / What is a neurotransmitter, really?

3 **Dopamine isn't 'pleasure'**
What dopamine actually signals — and why the pop-science version is wrong.
Brian asks: So why do people say it's the happy chemical? / What happens in addiction?

4 **The hippocampus and memory**
Why damage there stops new memories but leaves old ones.
Brian asks: Where do old memories go? / Why do I remember my first phone number?

5 **Sleep and memory**
What the brain does with the day while you're asleep.
Brian asks: So pulling an all-nighter before an exam is...? / How much sleep is enough?

6 **What fMRI actually measures**
Blood, not neurons — and what that means for every 'brain lights up' headline.
Brian asks: So it's not reading thoughts? / How slow is it?

7 **Stress and cortisol**
What the stress response is for, and what it does when it never switches off.
Brian asks: Is stress ever good? / What does chronic stress do to the brain?

8 **Your journal club paper**
The paper you presented in Class 3 — this time for someone with no science background.
Brian asks: Why should I care? / What's the one picture I should remember?

Pick three you could explain tonight without looking anything up. Number 8 is the only one Brian has heard of — and he still didn't follow it.

PART 1 · 2 of 3

How to Teach It

주제마다 같은 순서로 , 3 분 . 랩 발표의 뼈대와 같아요 — 청중만 비전공자로 바뀐 것 .

FIVE MOVES · 3 MINUTES

1 HOOK Why should a non-scientist care? One sentence.

2 WHAT WE KNOW The idea, in plain words. No jargon yet.

3 HOW WE KNOW IT The experiment or the patient that showed it.

4 WHY IT MATTERS What changed in the field, or in the clinic.

5 WHAT'S STILL OPEN The part nobody has figured out.

TOOLBOX

Imagine ~
상상해 보세요 , ~

Think of it like ~
~ 라고 생각하면 돼요

The surprising part is ~
놀라운 건 ~ 예요

We know this because ~
이걸 아는 건 ~ 덕분이에요

The catch is ~
문제는 ~ 예요

We still don't know ~
~ 는 아직 몰라요

BRIAN ASKS · THE NON-SCIENTIST'S QUESTIONS

So what? · How do we know that's true? · Could that happen to me? · Why did nobody know this before? · Is this settled, or still argued about?

PART 1 · 3 of 3

Round Two — Thirty Seconds

세 주제 다 끝나면 , 하나씩 30 초로 다시 . 노트 없이 . Brian 이 다른 사람한테 설명할 수 있으면 통과 .

TOPIC 1

Did Brian get it? 1 2 3 4 5

Hardest question:

Word Inyoung was missing:

TOPIC 2

Did Brian get it? 1 2 3 4 5

Hardest question:

Word Inyoung was missing:

TOPIC 3

Did Brian get it? 1 2 3 4 5

Hardest question:

Word Inyoung was missing:

INYOUNG SAYS · Brian writes the good sentences here

FROM YOUR WORDS · 지난주 단어 **coherent** · **explicit**

PART 2 · 1 of 4

Why Neuroscience

왜 이 길이었는지 . 아직 한 번도 안 물어본 것 .

1. When did you first decide on neuroscience — was there a moment, or did it creep up on you?
2. Was there a person you admired who pushed you this way?
3. How did you end up in this lab, of all labs?
4. What did your family say?
5. If not neuroscience, what would you be doing right now?

FROM YOUR WORDS · 지난주 단어

hesitant

TOOLBOX

What drew me to ~ was ~
~ 에 끌린 건 ~ 때문이었어요

I didn't plan it — ~
계획한 건 아니었어요 — ~

I admired the way ~
~ 하는 방식이 멋져 보였어요

Looking back, the turning point was ~
돌아보면 전환점은 ~ 였어요

If it weren't this, I'd probably ~
이게 아니었으면 아마 ~

CAUGHT TODAY

PART 2 · 2 of 4

A Day at KIST

실험실의 하루 . 몇 시부터 몇 시까지 , 뭐가 제일 싫고 뭐가 제일 좋은지 .

1. Walk me through a normal day — from the moment you arrive to the moment you leave.
2. What do you do regularly that nobody outside a lab would guess?
3. What's the most boring part? And the part you secretly enjoy?
4. When was the last time an experiment just failed? What did you do that evening?
5. How much of the day is actual science, and how much is everything else?

FROM YOUR WORDS · 지난주 단어

resilient

TOOLBOX

A typical day runs from ~ to ~
보통 하루는 ~ 부터 ~ 까지예요

Most of the morning goes to ~
오전은 대부분 ~ 에 써요

The part nobody sees is ~
아무도 모르는 부분은 ~ 예요

Honestly, half of it is ~
솔직히 절반은 ~ 예요

When it fails, I usually ~
실패하면 보통 ~ 해요

CAUGHT TODAY

PART 2 · 3 of 4

The Lab, Off the Record

사람들 , 랩 미팅 때 공기 , 교수님 . 솔직하게 .

1. Who's in the lab, and who do you actually talk to?
2. What's the air like in lab meeting — relaxed, tense, sleepy?
3. Have you ever pretended to understand something in a meeting? What happened?
4. What's the most embarrassing thing that's happened to you in the lab so far?
5. What's one thing about your professor that surprised you?

FROM YOUR WORDS · 지난주 단어

candid

TOOLBOX

There are ~ of us, and ~
저희는 ~ 명이고 , ~

In lab meeting, the mood is usually ~
랩 미팅 분위기는 보통 ~

I'll admit I once pretended to ~
솔직히 한 번은 ~ 하는 척했어요

The part I didn't expect was ~
예상 못 했던 건 ~ 였어요

What I've learned about ~ is ~
~ 에 대해 알게 된 건 ~ 예요

CAUGHT TODAY

PART 2 · 4 of 4

After the Master's

석사 끝나면 . 그리고 1 년 전의 나한테 한마디 .

1. When the master's is done — PhD, a job, abroad? Where are you leaning right now?
2. What makes you anxious about that choice, and what makes you confident?
3. Who do you talk to when you're not sure?
4. Where would you like to be at thirty?
5. One line to yourself a year ago — what would you say?

FROM YOUR WORDS · 지난주 단어

pragmatic

TOOLBOX

I'm leaning toward ~, but ~
~ 쪽으로 기울고 있는데 , ~

What makes me anxious is ~
불안한 건 ~ 예요

I feel confident about ~ because ~
~ 는 자신 있어요 , ~ 라서요

By thirty, I'd like to ~
서른까지는 ~ 하고 싶어요

If I could tell her one thing, it'd be ~
한마디만 할 수 있다면 ~

CAUGHT TODAY

WRAP

Lines Worth Keeping

오늘 Inyoung 입에서 나온 것 중 L&S 03 으로 갈 것 .

THE LINES WE KEEP